NETWORK & SECURITY ENGINEER

Hassan Ismail.

CS & AI Student, South Valley National University

Working across two tracks: competitive programming in C++ (ECPC/ICPC) and network security. On the security side, I design and harden network infrastructure — firewall clusters, SD-WAN edges, site-to-site IPsec tunnels, and segmented DMZ architectures on FortiGate — then prove every path with a policy trace and a CLI ping.

> “Failure is a milestone for success.”

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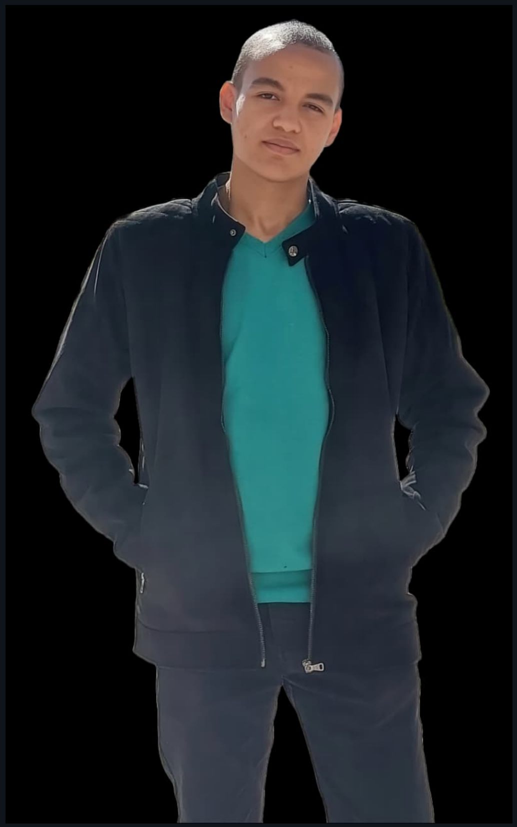
LinkedIn Profile

Qena, Egypt



ID: H.ISMAIL ● AVAILABLE

SELECTED WORK

01 **HA Cluster <-> SD-WAN Site-to-Site VPN**
Two FortiGate sites joined over an encrypted tunnel

02 **DMZ Segmentation**
Three-legged firewall isolating public-facing services

SKILLS & TOOLS

What's actually behind the work

Every item here is exercised somewhere in the two builds in this book.

CORE SKILLS

- Firewall Policy Design
- High Availability (HA) Clustering
- Site-to-Site IPsec VPN
- SD-WAN (Multi-WAN)
- Network Segmentation (DMZ)
- Static & Failsafe Routing
- CLI-Based Verification & Diagnostics

TOOLS & PLATFORMS

- FortiOS / FortiGate VM64
- FortiGate CLI Console
- Python (Intro + Intermediate)
- Network Fundamentals & Protocols

EDUCATION & TRAINING

Currently building the fundamentals

Second-year Computer Science & AI student, with applied network-security practice and two completed Python courses alongside coursework.

ENTERING YEAR 3

South Valley National University

Faculty of Computer Science & Artificial Intelligence - CS & IT

Completed the second year, now entering the third. Coursework in programming, algorithms, and computer networks runs alongside the independent FortiGate work in this book.

NTI - 2026

National Telecommunication Institute

FortiOS Administrator -- Network Security, 60-hour track

Official Fortinet Academy curriculum: firewall policies, the Security Fabric, authentication, and security profiles (IPS, antivirus, web filtering, app control), plus IPsec VPN, SD-WAN, and HA -- the practical basis for the FortiGate projects in this book, built toward the NSE4 FortiOS Administrator path (exam NSE4_FGT_AD-7.6). Also completed two Microsoft-backed Python courses under an MCIT digital-skills program.

Intro to Python
Sep 14-23, 2025

Python - Intermediate
Sep 25-Oct 2, 2025

COMPETITIVE PROGRAMMING

C++ contest record

Alongside the security work in this book, I compete in C++ with South Valley National University's ECPC/ICPC team.

APR 2025 - INDIVIDUAL

ICPC ACPC Kickoff Online Individual Contest

Honorable Mention

AUG 2025 - PERSONAL CERTIFICATE

ECPC Qualifications Collegiate Prog. Contest, Day 8

94th Place

AUG 2025 - TEAM CERTIFICATE

ECPC Qualifications Collegiate Prog. Contest, Day 8

94th Place

COMPETITIVE PROGRAMMING -- CONTINUED

AUG 2026 - PERSONAL CERTIFICATE

ECPC Qualifications
Collegiate Prog. Contest, Day 8

93rd Place


Certificate of Achievement

icpc.foundation

icpc International Collegiate Programming Contest

awarded to

Hassan Ismail Hassan Mohamed

South Valley National University

The 2026 ICPC ECPC Qualifications Collegiate Programming Contest Day 8
20 August 2026

Ninety-third Place

William B. Poucher, Ph. D.
ICPC Executive Director

AUG 2026 - TEAM CERTIFICATE

ECPC Qualifications
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93rd Place


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The 2026 ICPC ECPC Qualifications Collegiate Programming Contest Day 8
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Ninety-third Place

South Valley National University

Hisham Ali Mohamed Elsadek
Hassan Ismail Hassan Mohamed
Mostafa mohammed abo el hassan
Mohamed Hassan Esmael, Co-coach

William B. Poucher, Ph. D.
ICPC Executive Director

WHAT I DO

Services

Scope drawn directly from the two builds in this book.

01

Firewall & Policy Design

Rule sets built around least privilege - explicit deny where it matters, scoped ACCEPT rules everywhere else, all under UTM inspection.

Project 01 - 02

02

High Availability

Active-passive FortiGate clustering with unit priority and synchronization, so a single hardware failure doesn't take the network down.

Project 01

03

Site-to-Site VPN

IPsec tunnels between independent sites, with routing that prefers the tunnel and falls back to a blackhole route if it drops.

Project 01

04

SD-WAN

Multi-WAN edges grouped into a single zone, so outbound traffic keeps moving even if one link degrades.

Project 01

05

Network Segmentation

Zone-based design - LAN, DMZ, WAN - with firewall policy controlling what can reach what.

Project 02

06

Verification & Testing

Every path is confirmed from the CLI - sourced pings, tunnel status, session counters - before it's called done.

Project 01 - 02

PROJECT 01

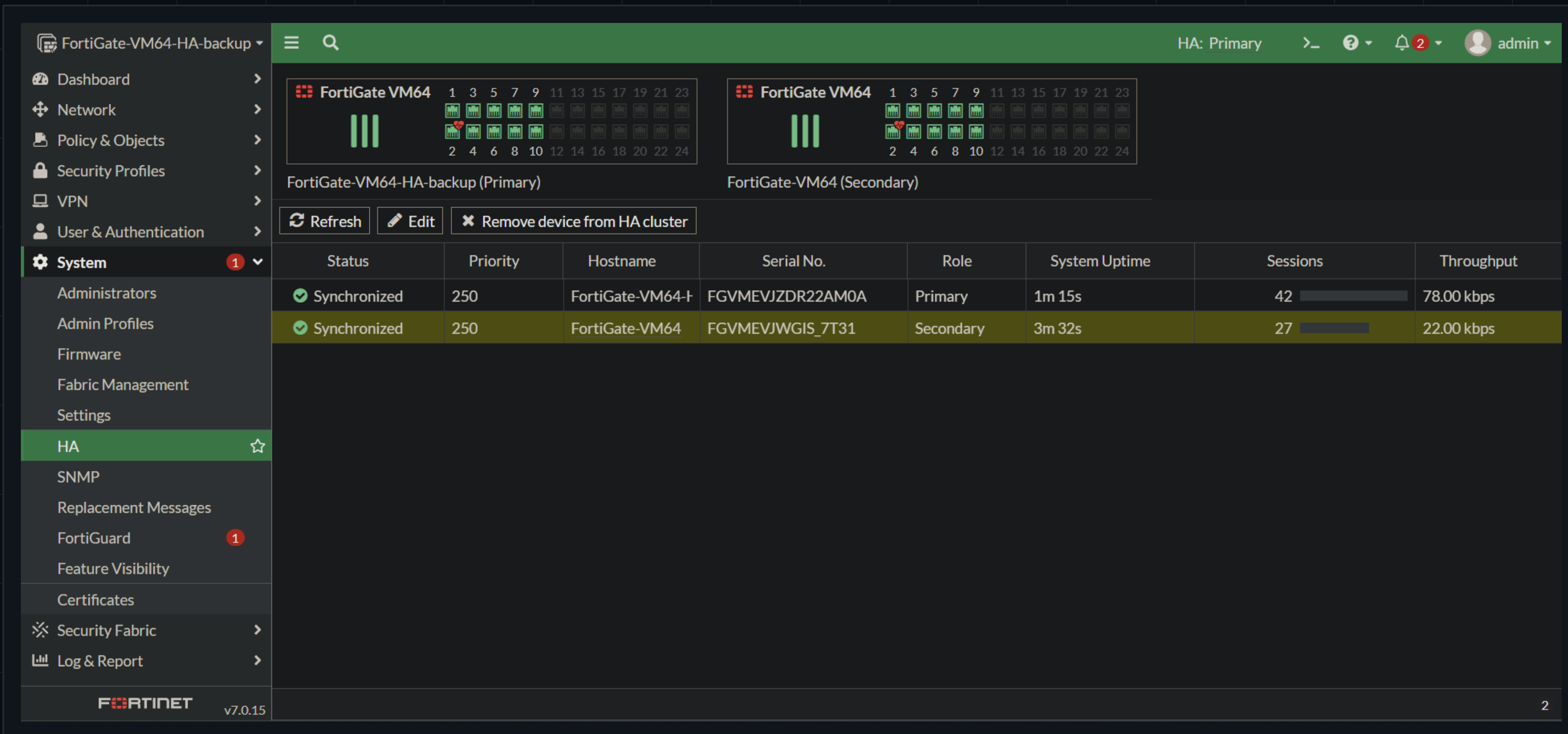
HA Cluster <-> SD-WAN Site-to-Site VPN

Two independent FortiGate deployments, built and validated as a single connected system: an active-passive HA pair at Site P1, and a dual-WAN SD-WAN edge at Site P2, joined by an IPsec tunnel and reachable subnet-to-subnet. Static routes on both sides prefer the tunnel and fall back to a blackhole route so traffic never leaks to the internet if the VPN drops.

`Role: Primary / Secondary` `Priority: 250` `Tunnel: Up - both directions` `Verification: CLI ping, 0% loss`

SITE P1 -- HA CLUSTER

Role: Primary/Secondary | Priority: 250 | LAN: 192.168.10.0/24 | Tunnel: vpn to p2 - Up

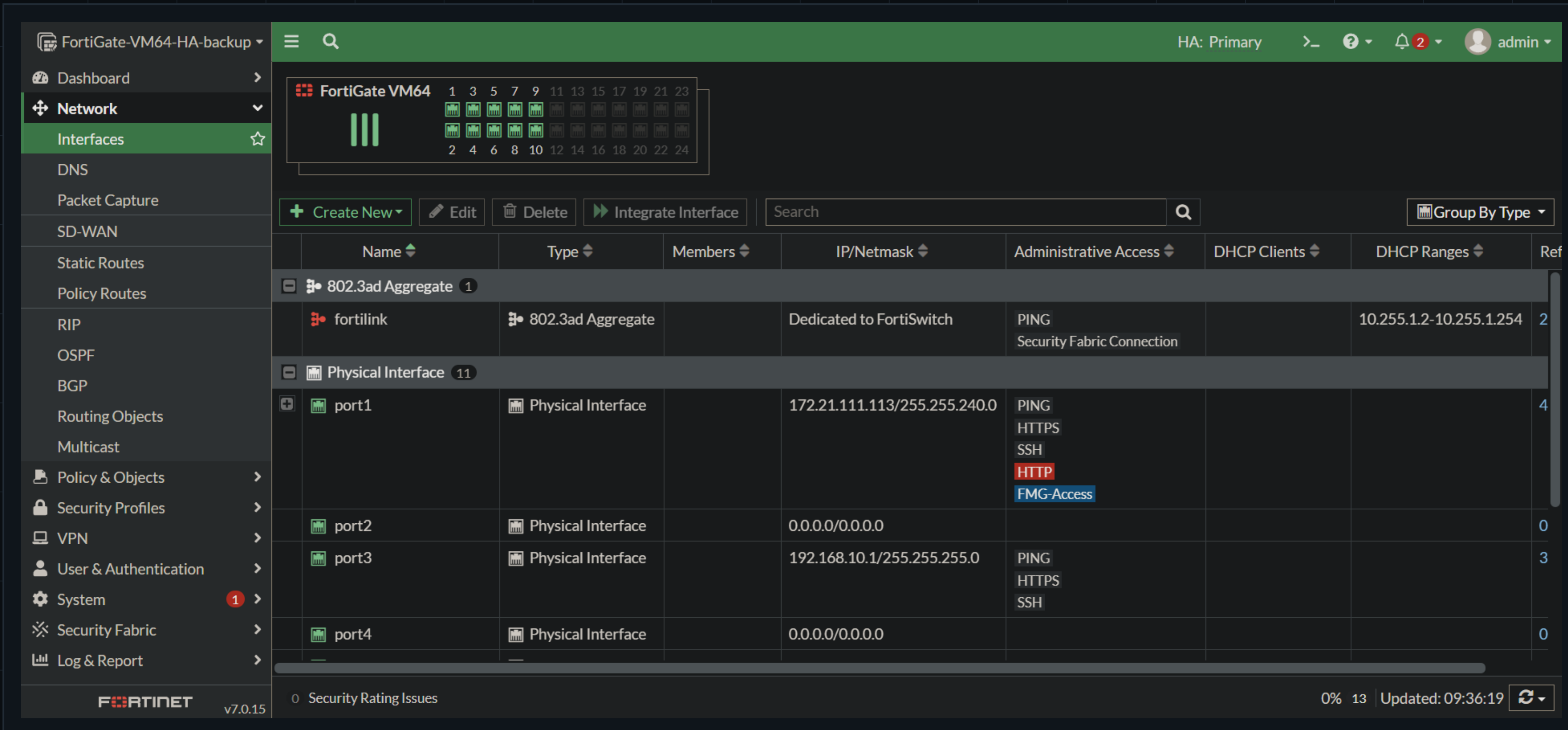


01 HA Status

Active-passive HA pair, both synchronized at priority 250 - live session count and throughput per unit.

● SITE P1 -- HA CLUSTER

Role: Primary/Secondary | Priority: 250 | LAN: 192.168.10.0/24 | Tunnel: vpn to p2 - Up

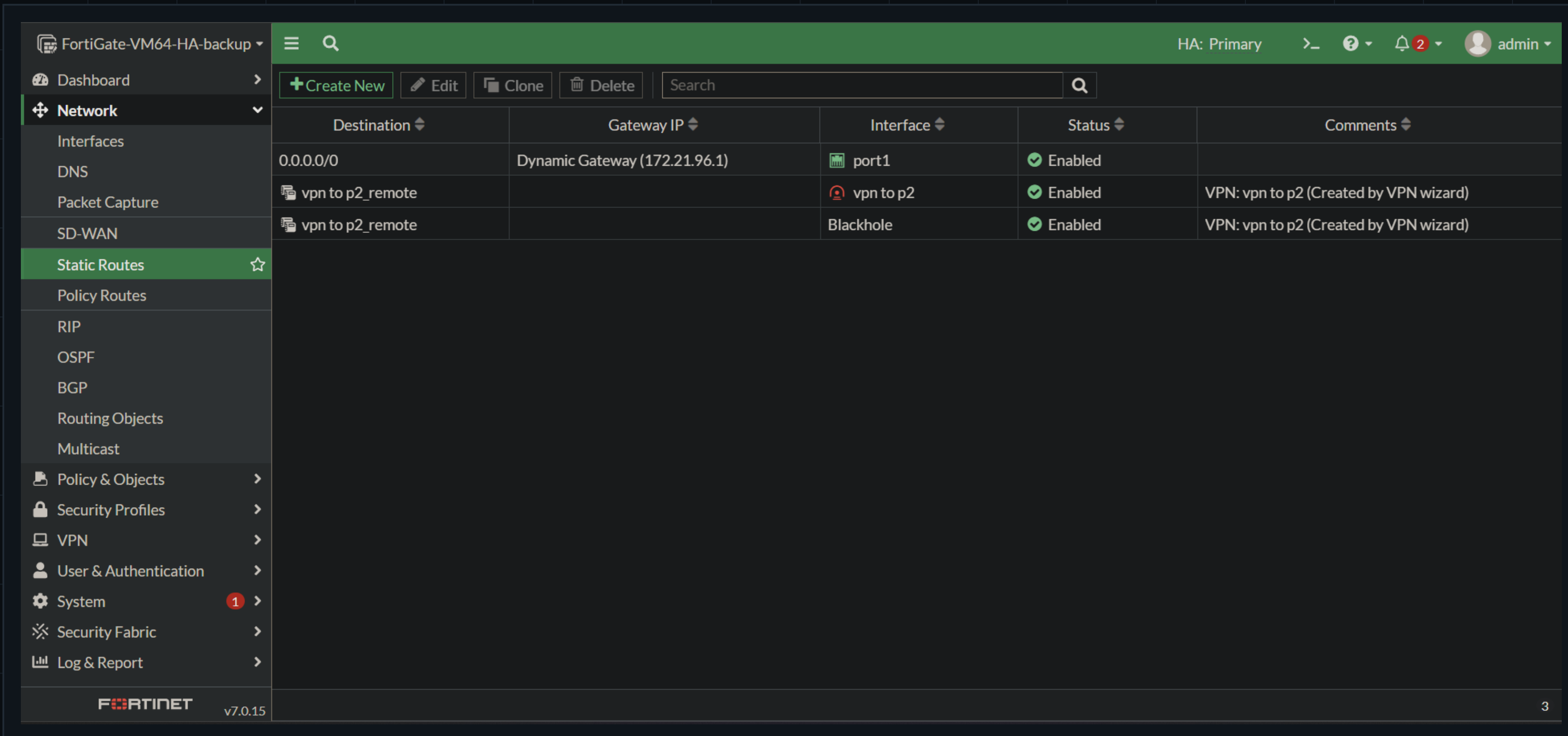


02 Interfaces

port1 carries WAN/management (SSH, HTTPS, FMG-Access); port3 terminates the internal 192.168.10.0/24 LAN.

SITE P1 -- HA CLUSTER

Role: Primary/Secondary | Priority: 250 | LAN: 192.168.10.0/24 | Tunnel: vpn to p2 - Up

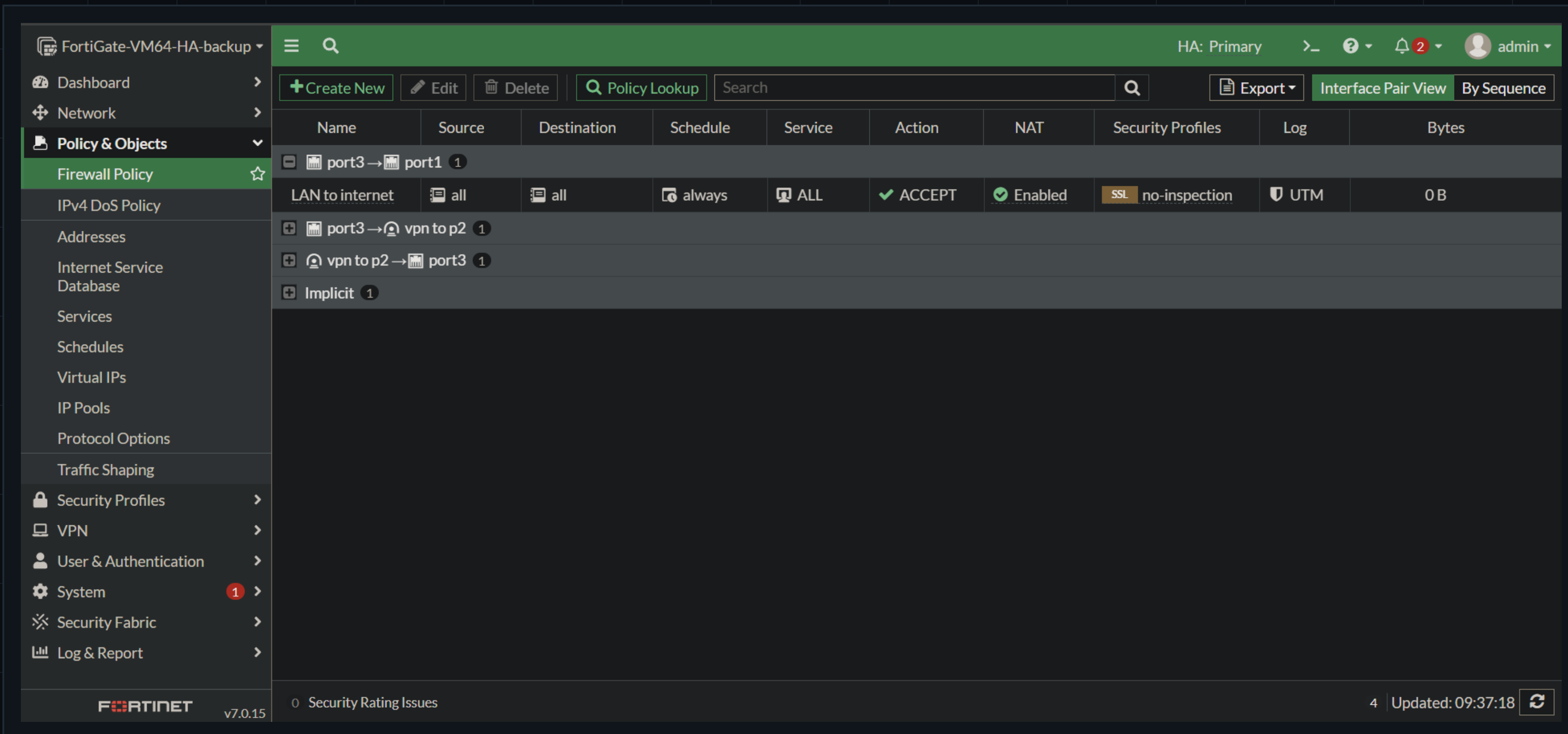


03 Static Routing

Default route out port1, a primary route through the tunnel, and a blackhole route so nothing leaks if the VPN drops.

● SITE P1 -- HA CLUSTER

Role: Primary/Secondary | Priority: 250 | LAN: 192.168.10.0/24 | Tunnel: vpn to p2 - Up



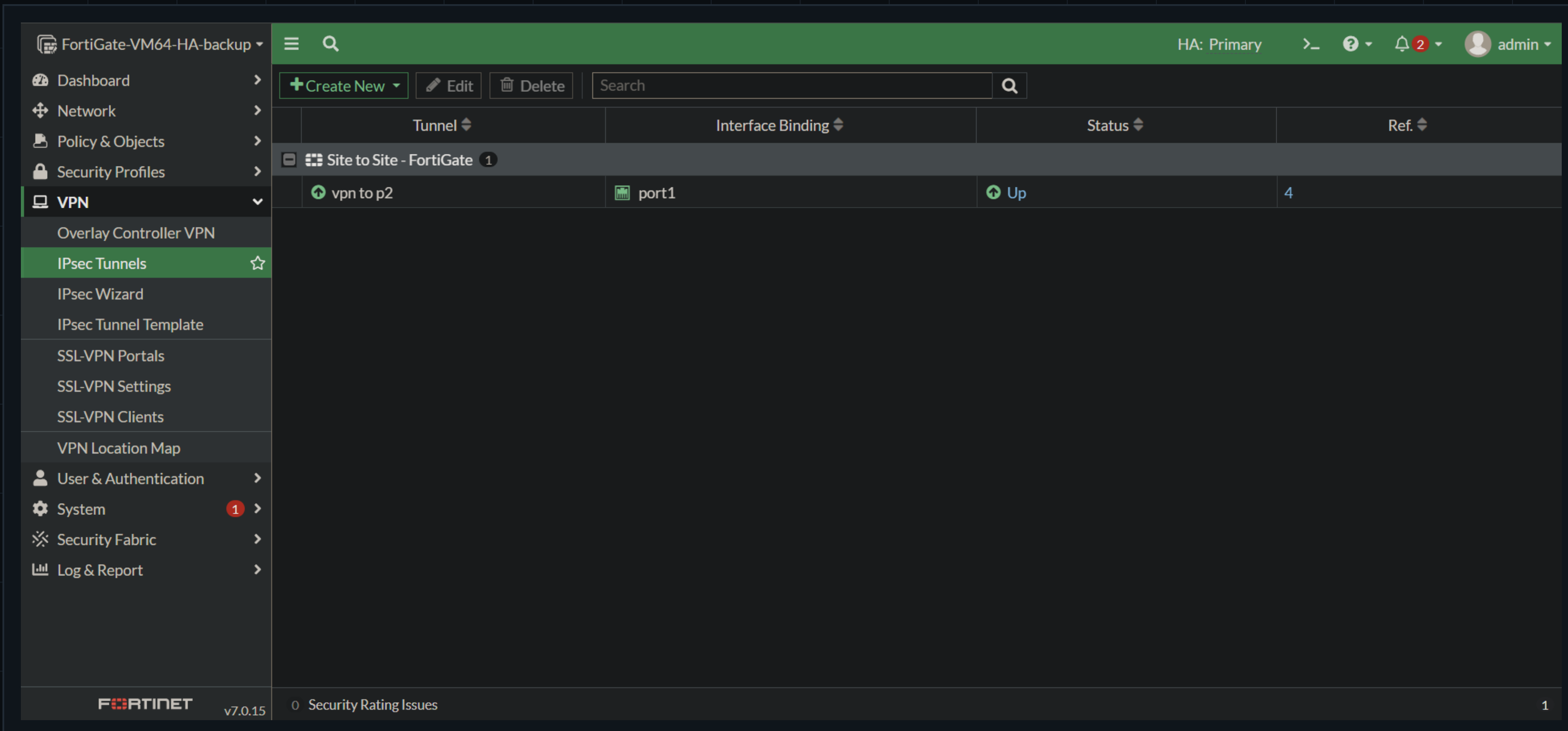
04 Firewall Policy

LAN-to-internet plus dedicated inbound/outbound rules for the VPN interface, all under UTM inspection.

● SITE P1 -- HA CLUSTER

PROJECT 01

Role: Primary/Secondary | Priority: 250 | LAN: 192.168.10.0/24 | Tunnel: vpn to p2 - Up

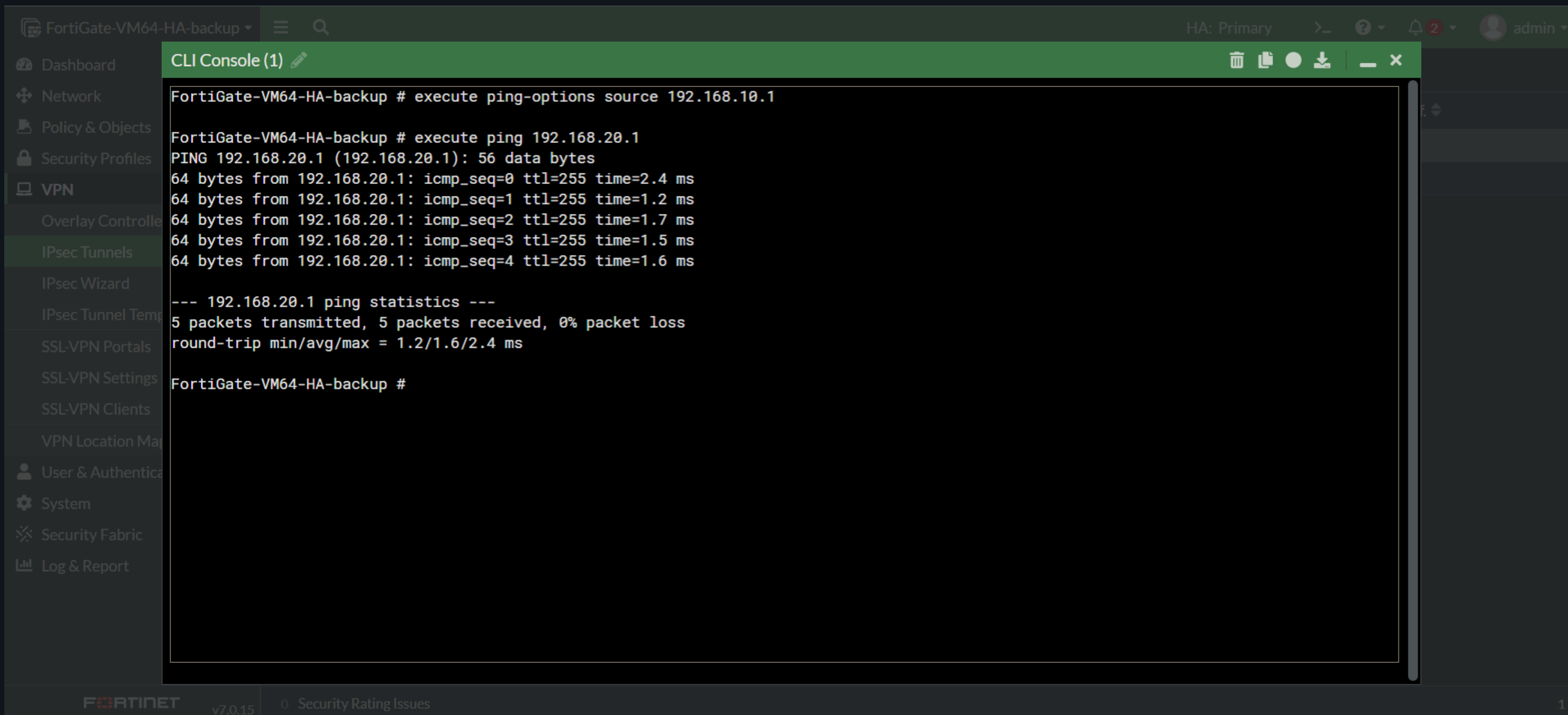


05 IPsec Tunnel

Site-to-site tunnel "vpn to p2," bound to port1, status Up.

SITE P1 -- HA CLUSTER

Role: Primary/Secondary	Priority: 250	LAN: 192.168.10.0/24	Tunnel: vpn to p2 - Up



06 CLI Verification

Sourced ping from the P1 LAN to the P2 LAN (192.168.20.1) - 5/5 packets, 0% loss, 1.6 ms average.

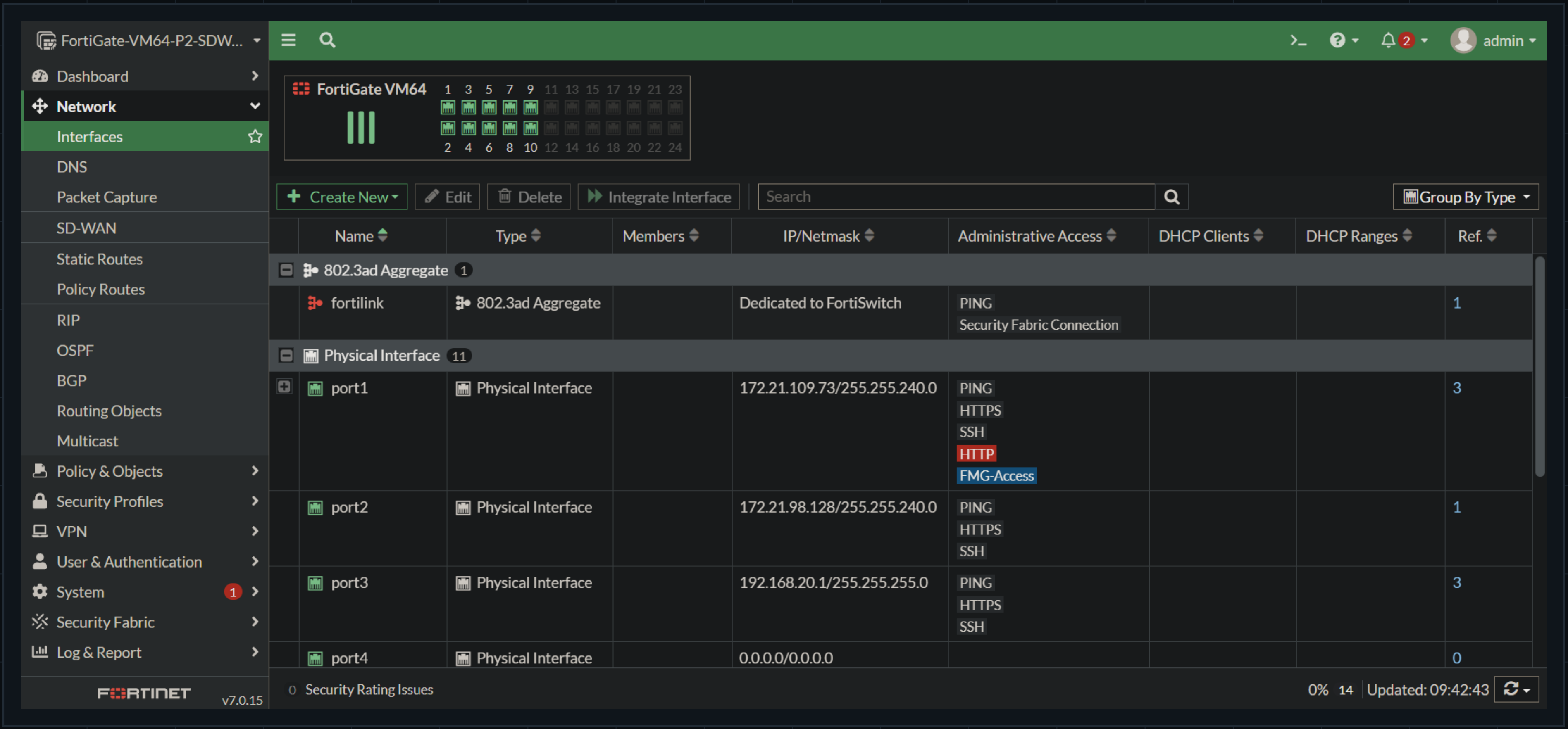
SITE P2 -- SD-WAN

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up



01 Interfaces

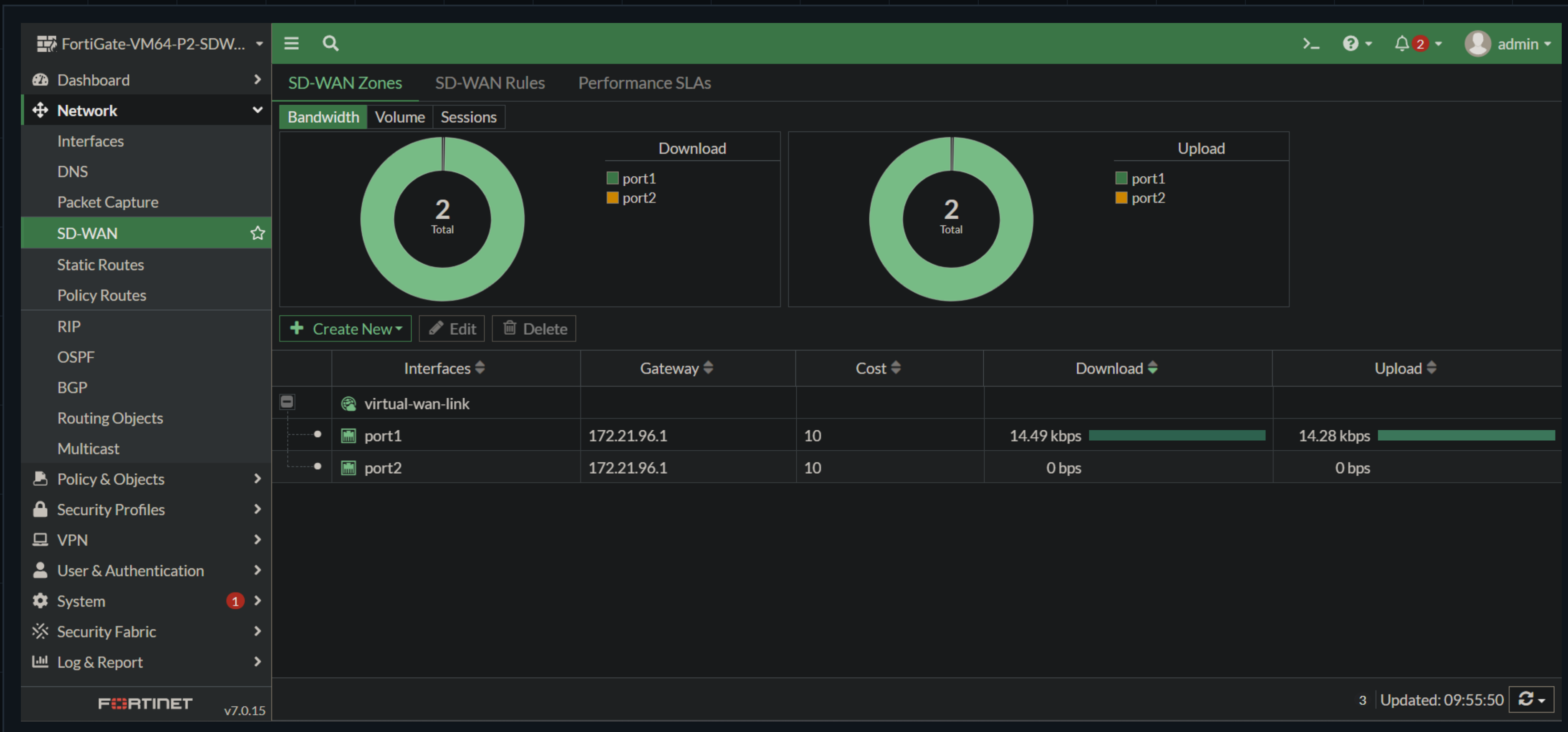
Two independent WAN links (port1, port2) plus the internal 192.168.20.0/24 LAN on port3.

● HASSAN ISMAIL

● SITE P2 -- SD-WAN

PROJECT 01

WAN links: port1 + port2 | SD-WAN zone: virtual-wan-link | LAN: 192.168.20.0/24 | Tunnel: vpn to p1 - Up



02 SD-WAN Zone

Both WAN ports grouped into the virtual-wan-link at equal cost, with live download/upload throughput per member.

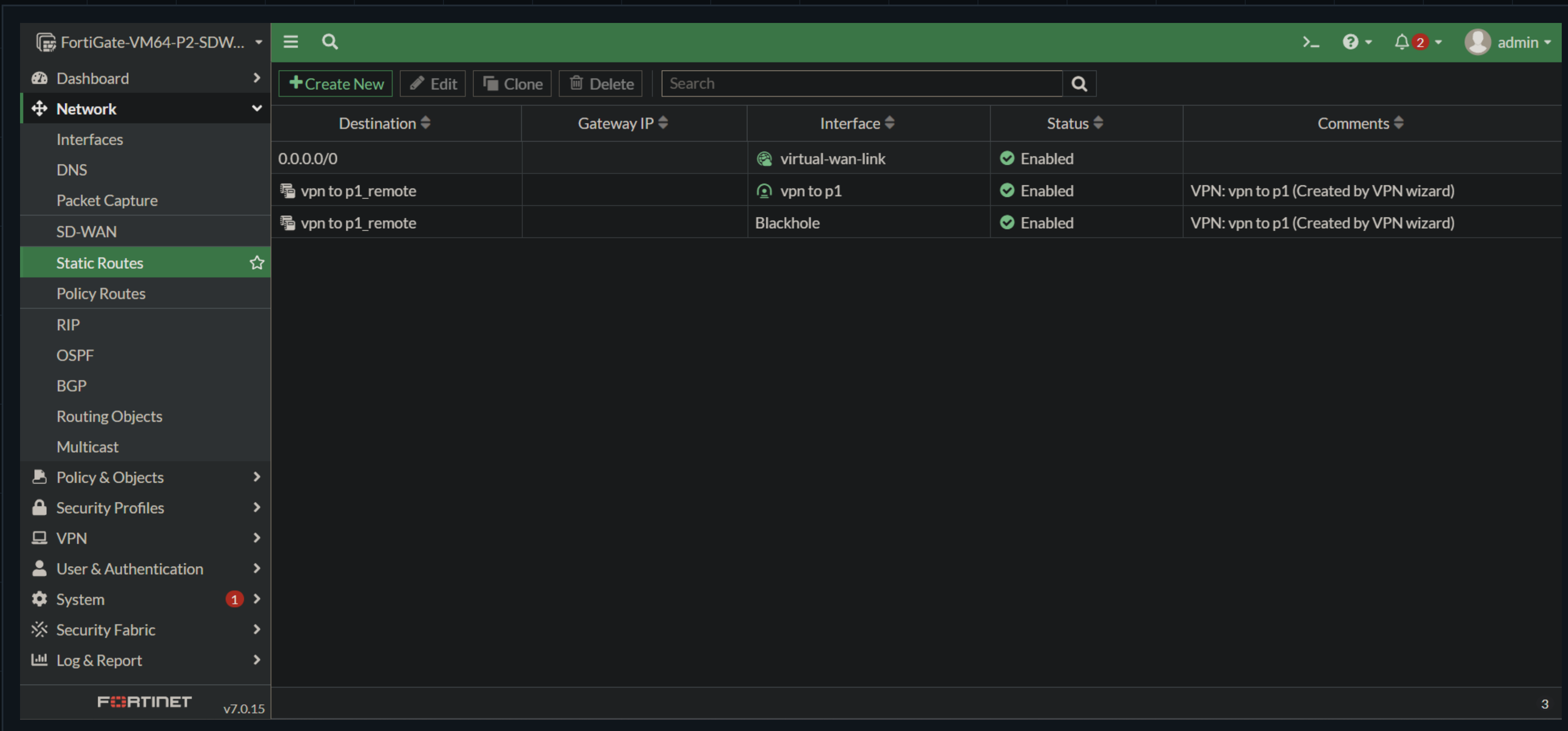
SITE P2 -- SD-WAN

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

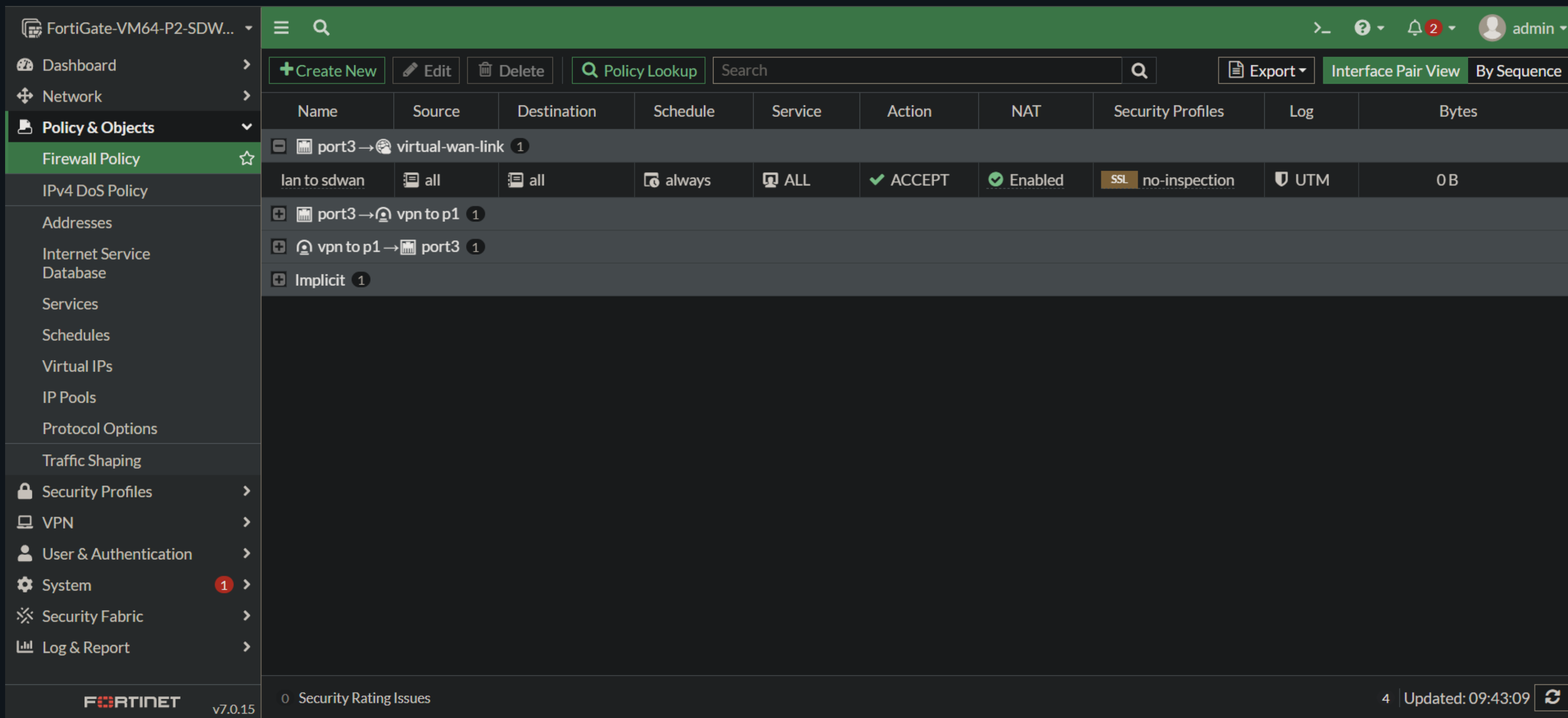


03 Static Routing

Default route rides the SD-WAN zone; the route to P1 gets the same primary-plus-blackhole failsafe pattern as P1.

SITE P2 -- SD-WAN

WAN links: port1 + port2 | SD-WAN zone: virtual-wan-link | LAN: 192.168.20.0/24 | Tunnel: vpn to p1 - Up



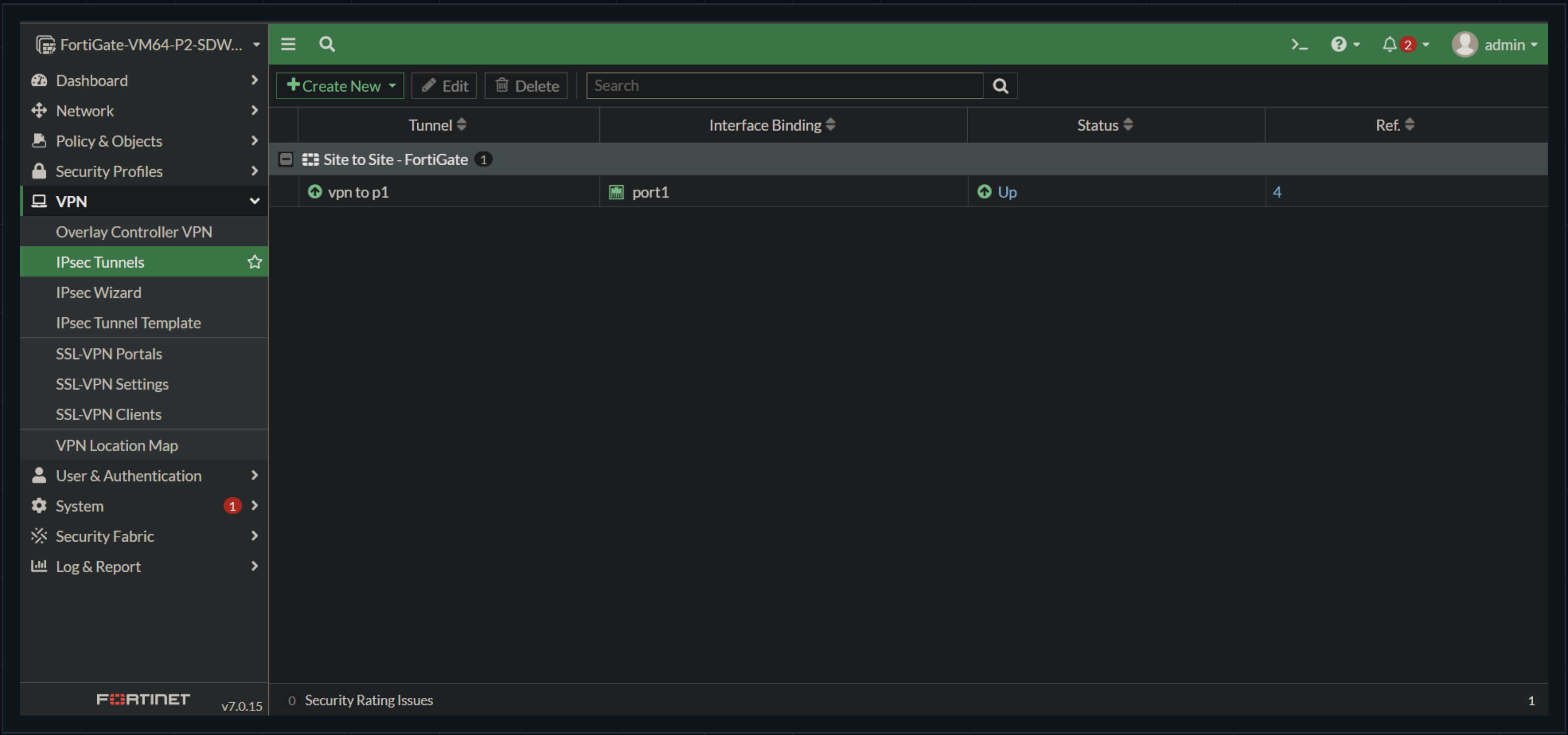
04 Firewall Policy

LAN-to-SD-WAN policy for general egress, mirrored by inbound/outbound rules for the tunnel back to P1.

● SITE P2 -- SD-WAN

PROJECT 01

WAN links: port1 + port2 | SD-WAN zone: virtual-wan-link | LAN: 192.168.20.0/24 | Tunnel: vpn to p1 - Up



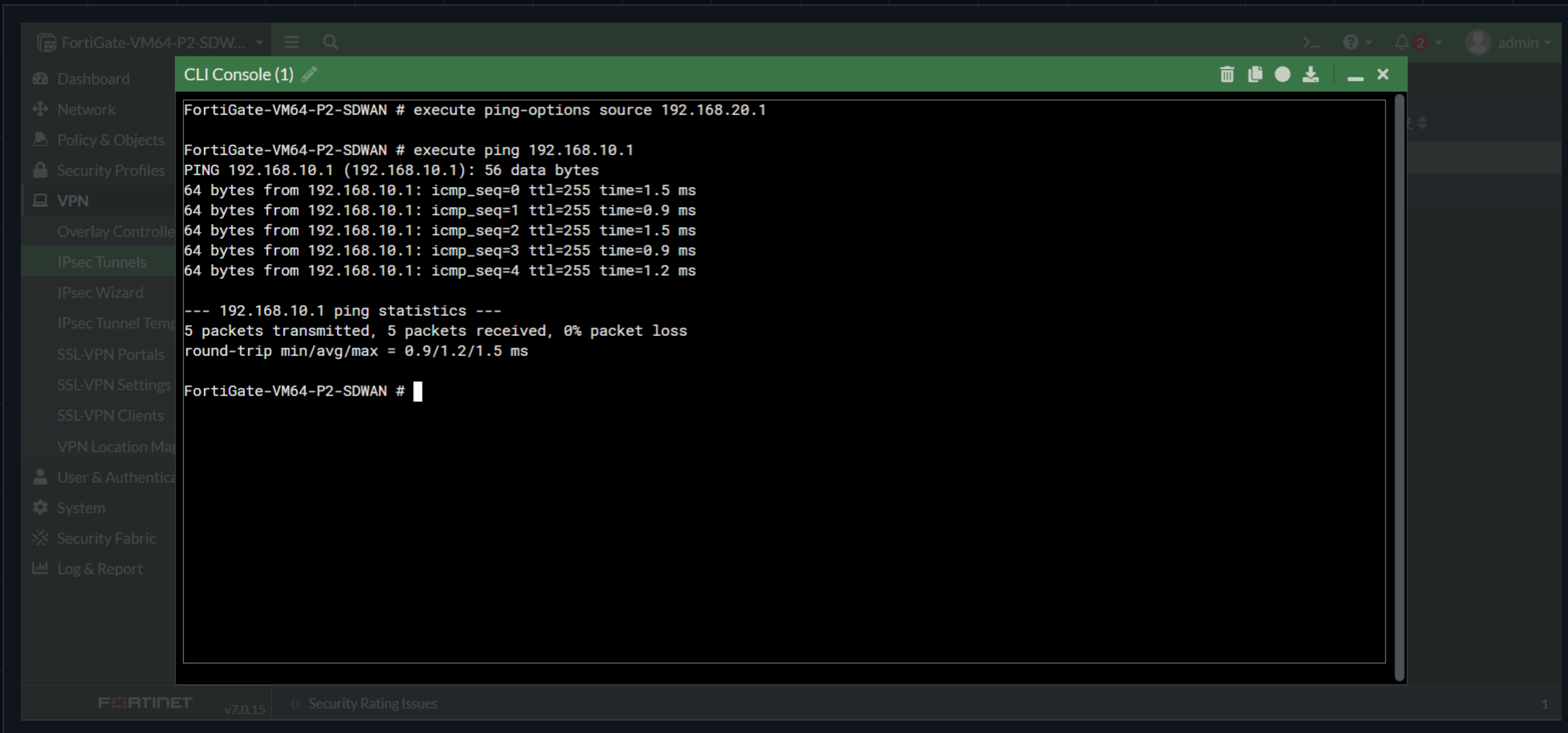
05 IPsec Tunnel

Site-to-site tunnel "vpn to p1," bound to port1, status Up.

SITE P2 -- SD-WAN

PROJECT 01

WAN links: port1 + port2 | SD-WAN zone: virtual-wan-link | LAN: 192.168.20.0/24 | Tunnel: vpn to p1 - Up



06 CLI Verification

Sourced ping from the P2 LAN to the P1 LAN (192.168.10.1) - 5/5 packets, 0% loss, 1.2 ms average, confirming bidirectional reachability.

PROJECT 02

DMZ Segmentation

A single FortiGate split into three security zones, so a compromised public-facing host in the DMZ can never pivot into the internal LAN. port1 faces the WAN, port2 hosts the internal LAN (192.168.10.0/24), and port3 is a dedicated DMZ (192.168.20.0/24) for exposed services. Policy does the isolating: DMZ-to-LAN traffic is explicitly denied, LAN-to-DMZ and LAN-to-WAN are permitted, and the implicit deny rule closes everything not named.

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted

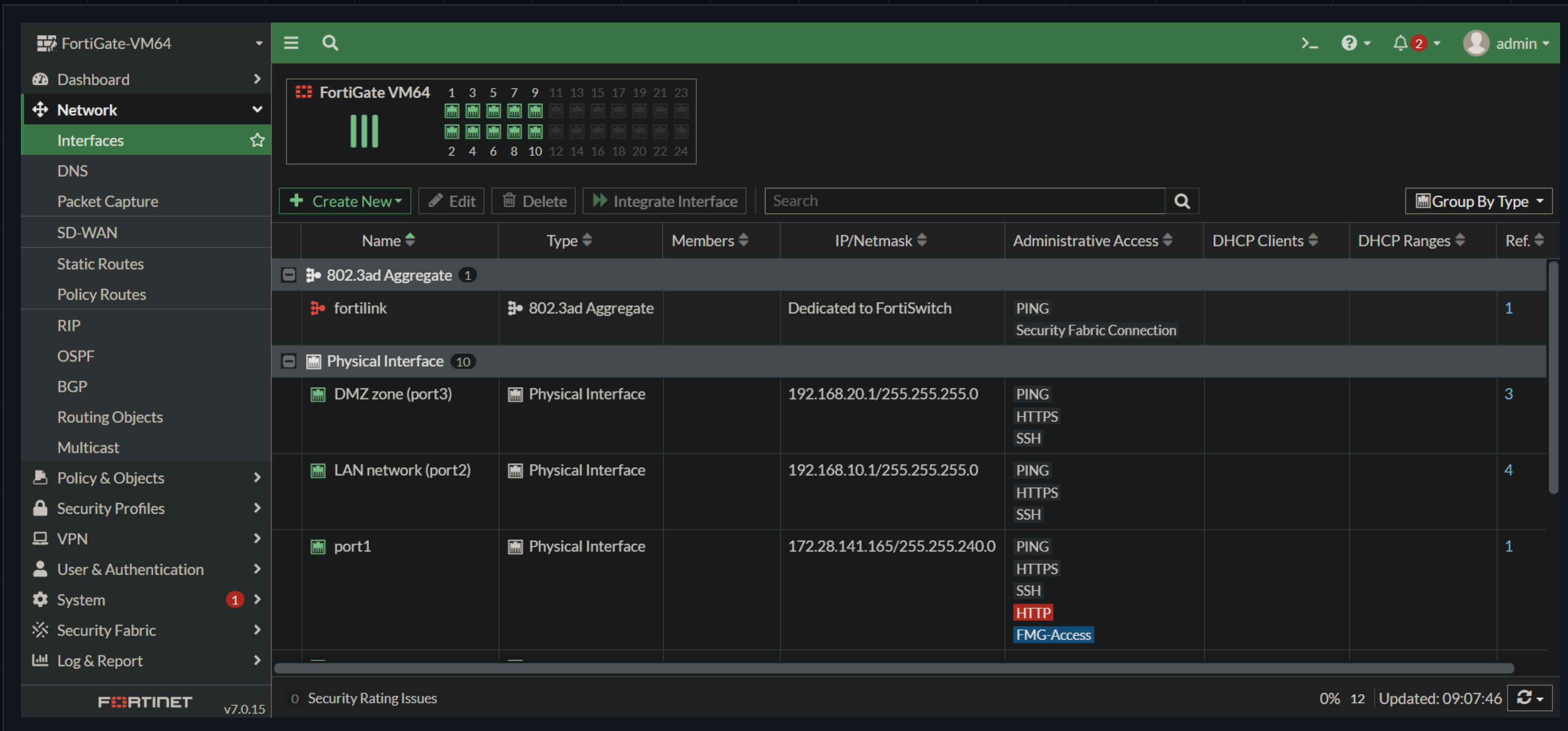
LAN -> WAN: Accepted

DMZ SEGMENTATION

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted



01 Interfaces

Dedicated DMZ (port3), internal LAN (port2), and WAN (port1) on a single FortiGate - each its own subnet.

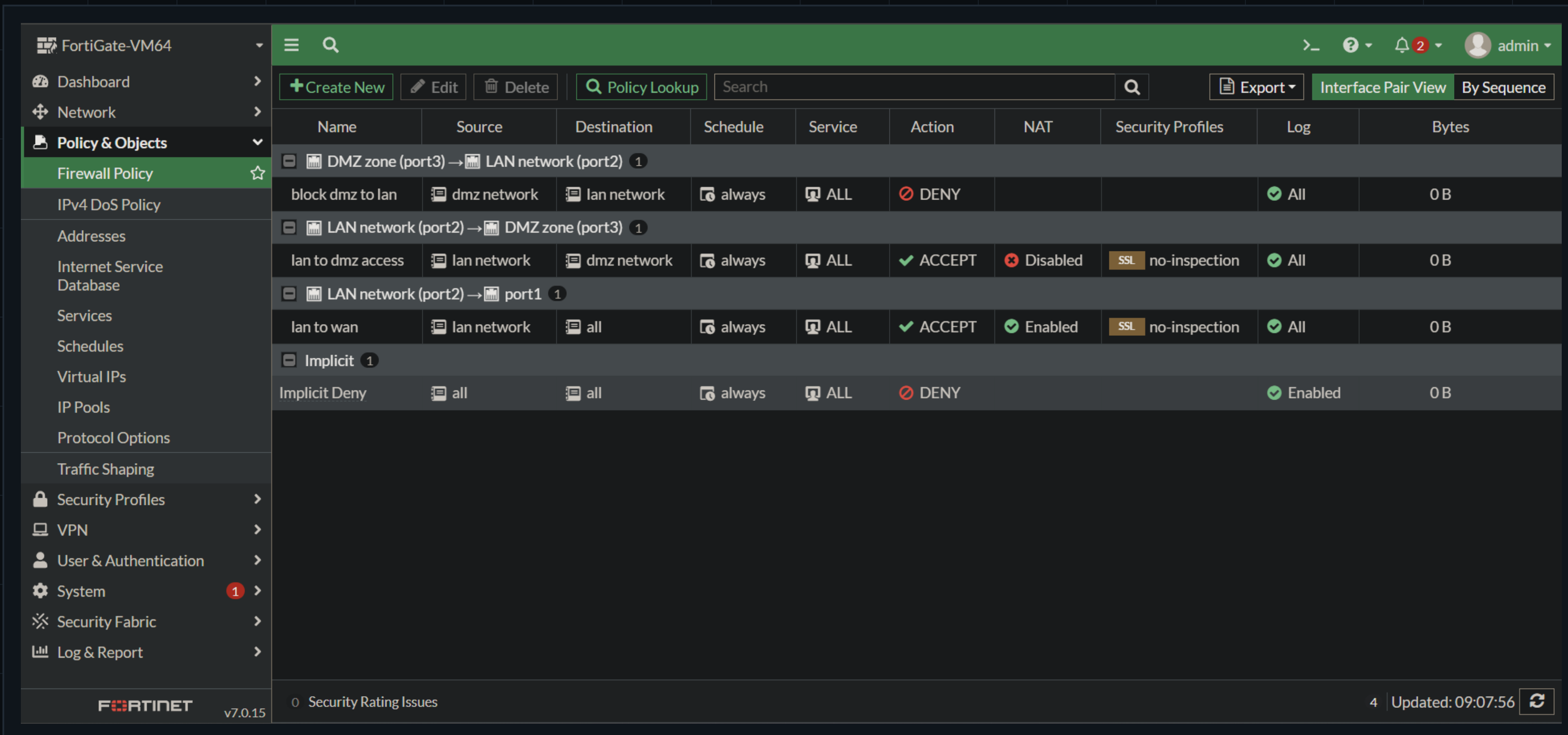
DMZ SEGMENTATION

PROJECT 02

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted



02 Firewall Policy

DMZ-to-LAN explicitly denied, LAN-to-DMZ and LAN-to-WAN permitted, implicit deny closes everything else.

HASSAN ISMAIL

GET IN TOUCH

Open to network & security engineering work.

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